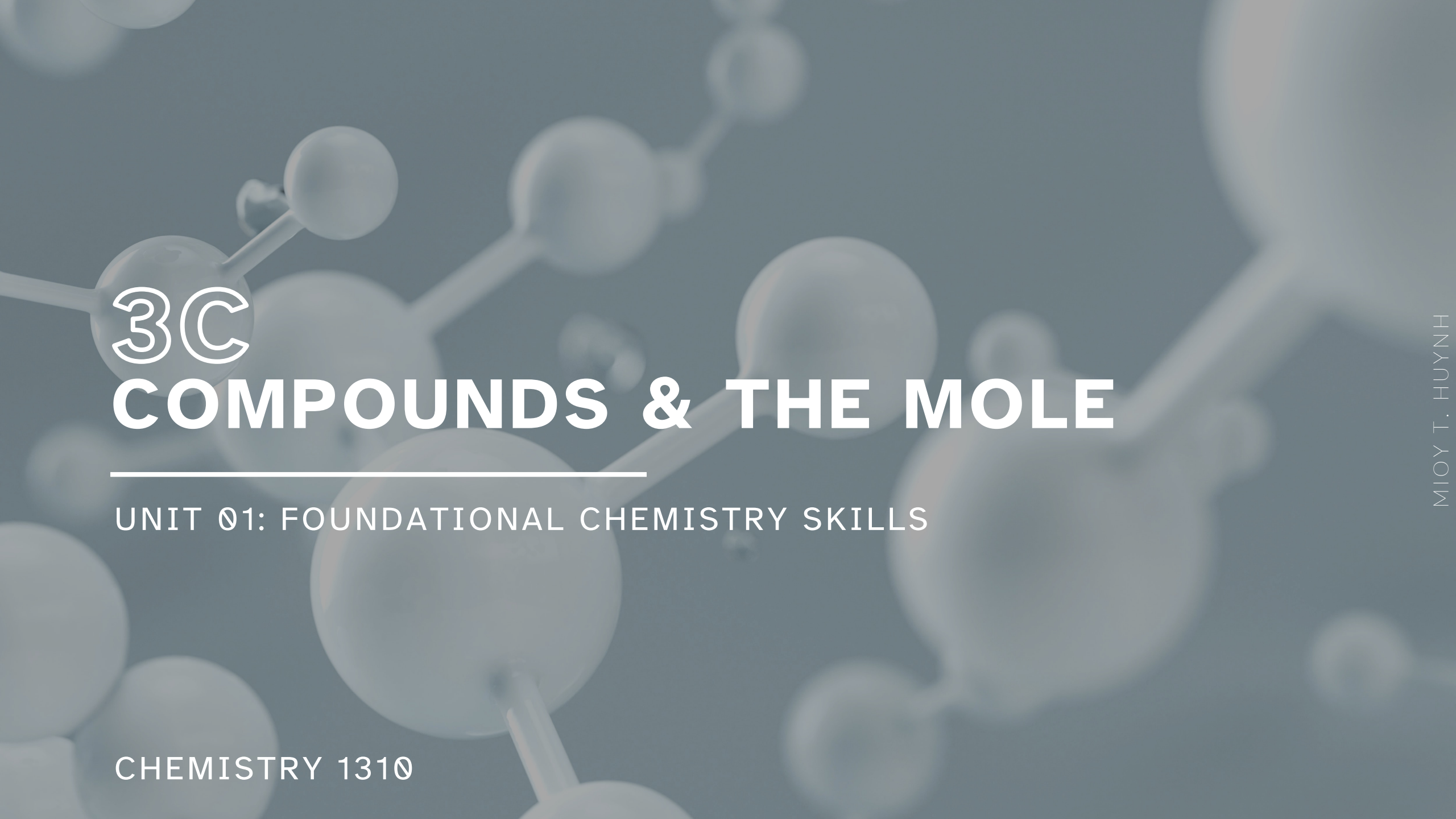


A faint, stylized molecular structure with spheres and connecting lines is visible in the background.

3C

COMPOUNDS & THE MOLE

UNIT 01: FOUNDATIONAL CHEMISTRY SKILLS

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 3

1. Interpret chemical formulas.
2. Write chemical formulas to represent simple inorganic chemical compounds.
3. Name simple inorganic chemical compounds.
4. Calculate the number of moles from the number of particles and vice versa.
5. Calculate the mass of 1 mole of a substance based on its chemical formula and atomic masses from the periodic table.
6. Calculate the percent composition by mass of a compound based on its chemical formula and atomic masses from the periodic table.
7. Determine the empirical and molecular formulas of a compound from percent composition or other mass-ratio data.
8. Determine the empirical formula for a compound based on data from a combustion analysis.

Molar Mass

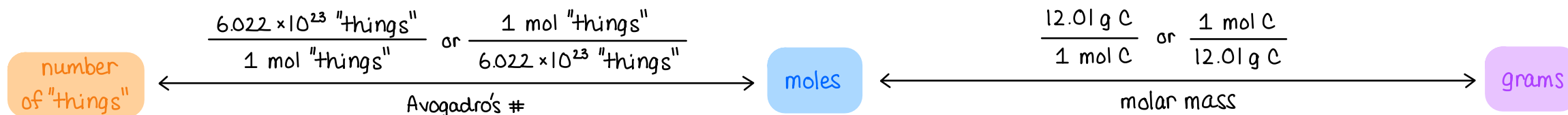
3.8

Atomic mass: For an element, the atomic mass is the average mass of one atom, expressed in atomic mass units (amu).

- One atomic mass unit, amu, is equal to 1 gram (g) divided by Avogadro's number such that: $1 \text{ amu} = 1.661 \times 10^{-24} \text{ g}$

Molar mass: For an element, the molar mass is the mass in grams of exactly 1 mole (or 6.022×10^{23} atoms) of that element

- Molar mass allows conversion between mass and the number of moles.

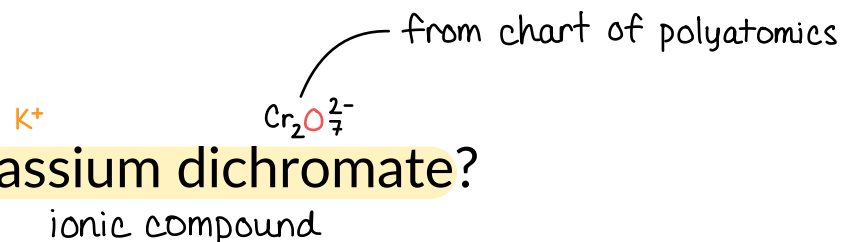


"things" = atoms, molecules, formula units, etc.

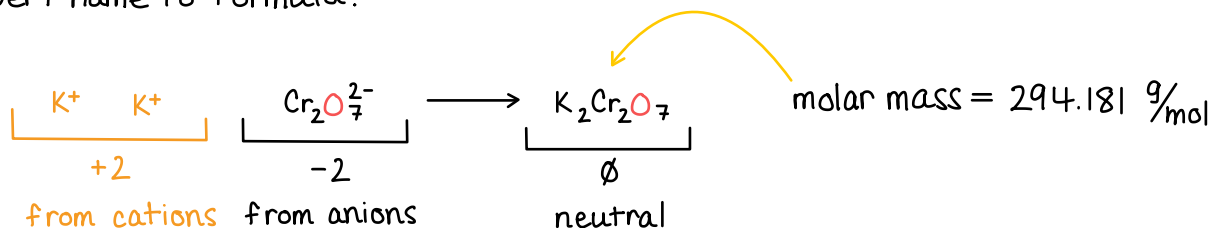
IN-CLASS QUESTION 1

Learning Objective(s): 3.4, 3.5

How many atoms of oxygen are present in 255 g of potassium dichromate?



Convert name to formula.



1 formula unit $K_2Cr_2O_7$: 2 atoms K : 2 atoms Cr : 7 atoms O

1 mol $K_2Cr_2O_7$: 2 mol K : 2 mol Cr : 7 mol O

General Strategy: mass $K_2Cr_2O_7$ $\xrightarrow{\text{molar mass}}$ mol $K_2Cr_2O_7$ $\xrightarrow{\text{chemical formula ratio}}$ mol O $\xrightarrow{\text{Avogadro}}$ atoms O

$$255 \text{ g } K_2Cr_2O_7 \times \frac{1 \text{ mol } K_2Cr_2O_7}{294.181 \text{ g } K_2Cr_2O_7} = 0.8668 \text{ mol } K_2Cr_2O_7 \times \frac{7 \text{ mol O}}{1 \text{ mol } K_2Cr_2O_7} = 6.067 \text{ mol O} \times \frac{6.022 \times 10^{23} \text{ atoms O}}{1 \text{ mol O}} = 3.65 \times 10^{24} \text{ atoms O}$$

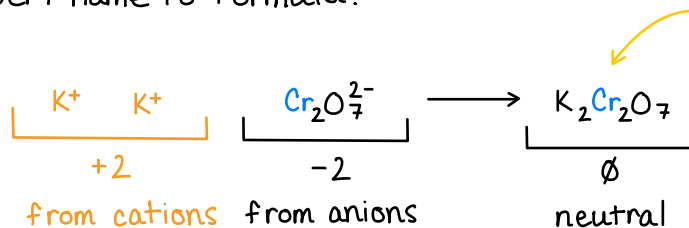
IN-CLASS QUESTION 2

Learning Objective(s): 3.4, 3.5, 3.6

What is the mass of chromium in 255 g of potassium dichromate?

from chart of polyatomics
 K^+ $Cr_2O_7^{2-}$
 ionic compound

Convert name to formula.



molar mass = 294.181 g/mol

1 formula unit $K_2Cr_2O_7$: 2 atoms K : 2 atoms Cr : 7 atoms O

1 mol $K_2Cr_2O_7$: 2 mol K : 2 mol Cr : 7 mol O

General Strategy: mass $K_2Cr_2O_7$ $\xrightarrow{\text{molar mass}}$ mol $K_2Cr_2O_7$ $\xrightarrow{\text{chemical formula ratio}}$ mol Cr $\xrightarrow{\text{molar mass}}$ mass Cr

$$255 \text{ g } K_2Cr_2O_7 \times \frac{1 \text{ mol } K_2Cr_2O_7}{294.181 \text{ g } K_2Cr_2O_7} = 0.8668 \text{ mol } K_2Cr_2O_7 \times \frac{2 \text{ mol Cr}}{1 \text{ mol } K_2Cr_2O_7} = 1.733 \text{ mol Cr} \times \frac{51.996 \text{ g Cr}}{1 \text{ mol Cr}} = 90.1 \text{ g Cr}$$

Percent Composition

3.9

Percent composition tells us what percentage (by mass) of a compound is made up by each element in the compound.

- May be calculated using formula masses (atomic mass units, amu) but more often using molar masses (g/mol).

$$\% X = \frac{\text{total mass of all } X}{\text{total mass of compound}} \times 100\%$$


Consider methane, CH₄:

$$\text{CH}_4 : 12.01 + 4(1.008) = 16.04 \text{ g/mol}$$

$$\frac{12.01}{16.04} \times 100\% = 74.9\% \text{ C}$$

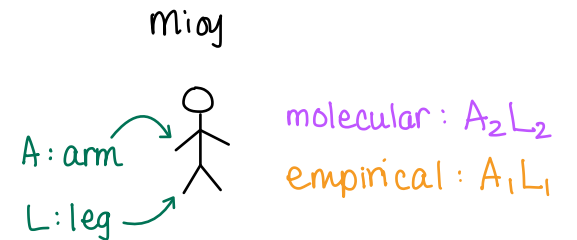
$$\frac{4 \times 1.008}{16.04} \times 100\% = 25.1\% \text{ H}$$

Percent Composition

3.9

Chemical formula: Describes compounds in terms of their atomic constituents

- Molecular formula "actual" formula
 - The complete inventory of atoms
 - How many of each element in the compound
- Empirical formula
 - The lowest whole number ratio of atoms in a compound



Chemical formulas can either represent atom-to-atom ratios or mole-to-mole ratios!

Empirical Formulas

3.10

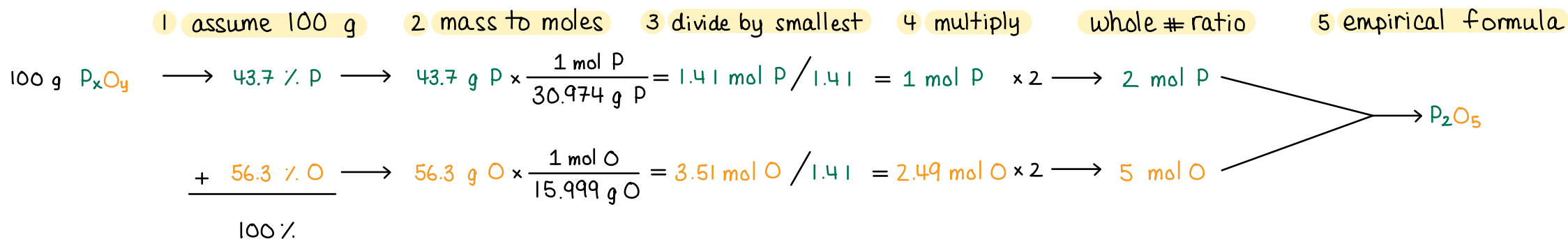
Steps in determining an empirical formula:

1. Convert mass percent data into masses by assuming a 100 g sample.
2. Convert the mass of each element into moles of each element using molar masses.
3. Divide moles of all elements by the smallest number of moles.
4. If not whole numbers, multiply by an appropriate factor.
5. Insert the numbers as subscripts in the chemical formula.

EXAMPLE PROBLEM 1

Learning Objective(s): 3.7

Determine the empirical formula of a compound that has a composition of 43.7 % P and 56.3 % O by mass.



Molecular Formulas

3.11

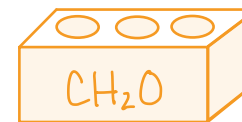
The molecular formula gives the actual number of atoms of each element in one *molecule* of a covalent compound.

- The molecular formula is always an integer (whole-number) *multiple* of the empirical formula.
- For example, one molecule of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) contains 6 C atoms, 12 H atoms, and 6 O atoms

empirical formula CH_2O

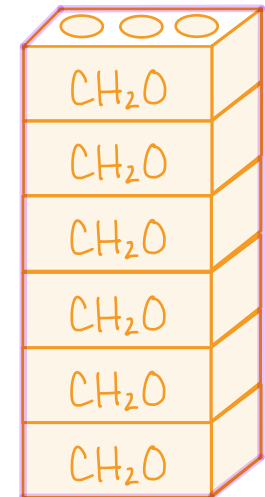
molecular formula $(\text{CH}_2\text{O})_n$ or $\text{C}_n\text{H}_{2n}\text{O}_n$

$$n = \frac{\text{molecular mass}}{\text{empirical mass}}$$



30.03 g

$\times 6 =$



$$30.03 \text{ g} \times 6 = 180.18 \text{ g}$$

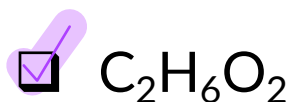
IN-CLASS QUESTION 3

Learning Objective(s): 3.7

A compound has the empirical formula CH_3O and a formula mass of 62.10 amu.

molecular formula

What is the molecular formula of the compound?

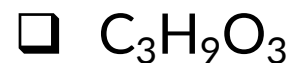


empirical : CH_3O

, empirical mass = 31.04 amu

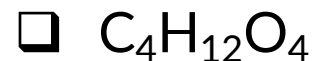


molecular : $(\text{CH}_3\text{O})_n$, molecular mass = 62.10 amu



$$n = \frac{\text{molecular mass}}{\text{empirical mass}} = \frac{62.10 \text{ amu}}{31.04 \text{ amu}}$$

$$n = 2$$



EXAMPLE PROBLEM 3

Learning Objective(s): 3.7, 3.8

A compound containing only carbon, hydrogen, and oxygen is found to be 41.39 % C and 3.47 % H by mass. If the molar mass of the compound is 116.28 g/mol, what is the molecular formula of the compound?

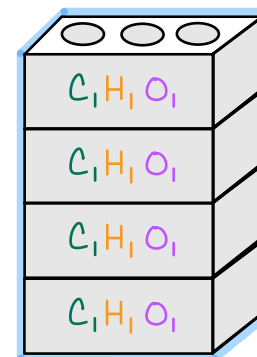
mass to moles divide by smallest

$$\begin{array}{l}
 \text{Assume } 100 \text{ g } C_xH_yO_z \longrightarrow 41.39\% \text{ C} \longrightarrow 41.39 \text{ g C} \times \frac{1 \text{ mol C}}{12.011 \text{ g C}} = 3.446 \text{ mol C} / 3.442 = 1 \text{ mol C} \\
 3.47\% \text{ H} \longrightarrow 3.47 \text{ g H} \times \frac{1 \text{ mol H}}{1.008 \text{ g H}} = 3.442 \text{ mol H} / 3.442 = 1 \text{ mol H} \\
 + \frac{55.14\% \text{ O}}{100\%} \longrightarrow 55.14 \text{ g O} \times \frac{1 \text{ mol O}}{15.999 \text{ g O}} = 3.446 \text{ mol O} / 3.442 = 1 \text{ mol O}
 \end{array}
 \longrightarrow \text{empirical formula} = C_1H_1O_1$$

empirical mass = 29.018 g/mol

molecular formula = $(C_1H_1O_1)_n = (C_1H_1O_1)_4 = C_4H_4O_4$

$$n = \frac{\text{molecular mass}}{\text{empirical mass}} = \frac{116.28 \text{ g/mol}}{29.018 \text{ g/mol}} = 4$$



Combustion Analysis

3.12

Combustion analysis is a lab technique used to determine empirical formulas of **hydrocarbon**, compounds containing *only* **C** and **H** (C_xH_y) – and perhaps **O** ($C_xH_yO_z$).

- A known mass of the unknown sample is reacted with oxygen (O_2).
- All **carbon** from the sample ends up as CO_2 .
- All **hydrogen** from the sample ends up as H_2O .
- The masses of CO_2 and H_2O are measured and used to determine mass of **C** and **H** in the original unknown sample.

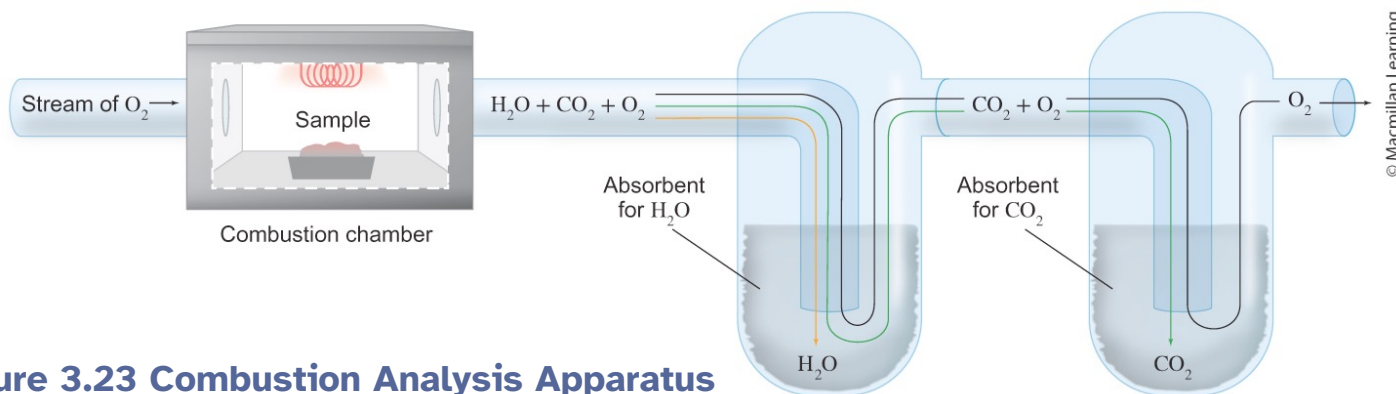
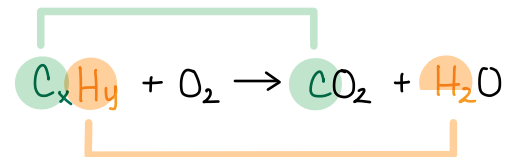


Figure 3.23 Combustion Analysis Apparatus

Combustion Analysis

3.12

Combustion analysis is a lab technique used to determine empirical formulas of **hydrocarbon**, compounds containing *only* **C** and **H** (C_xH_y) – and perhaps **O** ($C_xH_yO_z$).

- The masses of CO_2 and H_2O are measured and used to determine mass of **C** and **H** in the original unknown sample.

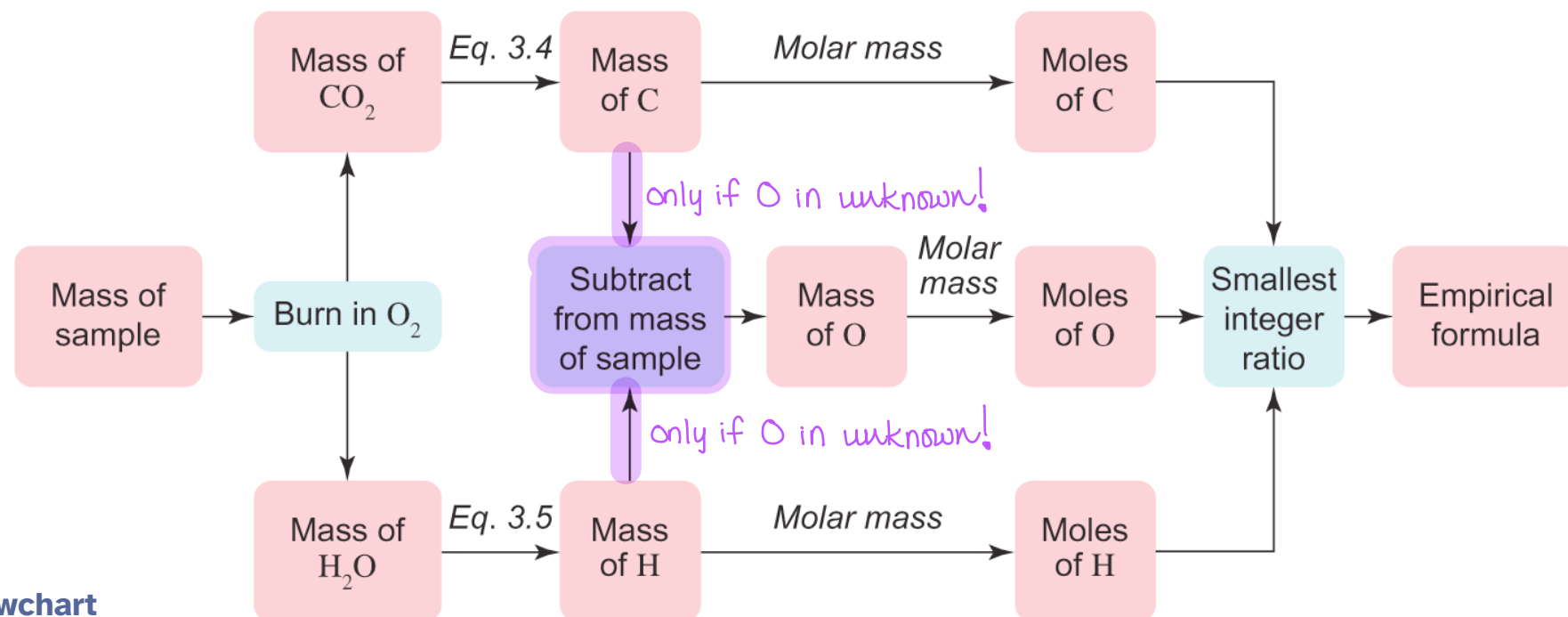
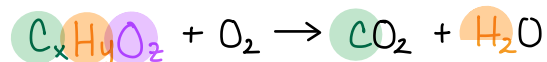
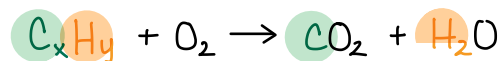


Figure 3.24 Combustion Analysis Flowchart

EXAMPLE PROBLEM 2

Learning Objective(s): 3.7, 3.8



A 0.250 g sample of a hydrocarbon, containing only C and H atoms, undergoes complete combustion to produce 0.845 g of CO_2 and 0.173 g of H_2O .

What is the empirical formula of this compound?

mass ——— MM ——— mole:mole

$$0.845 \text{ g CO}_2 \times \frac{1 \text{ mol CO}_2}{44.01 \text{ g CO}_2} \times \frac{1 \text{ mol C}}{1 \text{ mol CO}_2} = 0.0192 \text{ mol C} \div 0.0192 = 1 \text{ mol C}$$

$$0.173 \text{ g H}_2\text{O} \times \frac{1 \text{ mol H}_2\text{O}}{18.02 \text{ g H}_2\text{O}} \times \frac{2 \text{ mol H}}{1 \text{ mol H}_2\text{O}} = 0.0192 \text{ mol H} \div 0.0192 = 1 \text{ mol H}$$

CH

divide by smaller
of moles to get ratio

IN-CLASS QUESTION 4

multiple solutions!

Learning Objective(s): 3.5, 3.7

Vitamin C contains the elements C, H, and O. It is known to contain 40.9% C and 4.58% H by mass. The mass of 0.500 moles of vitamin C is 88.07 g. How many atoms of carbon are in a molecule of vitamin C?

Approach 1: Determine # mol C in 1 mole Vitamin C since chemical formula represents mole:mole = atom:atom ratios.

$$\begin{aligned}\text{MM Vit C} &= \frac{\text{mass}}{\text{mol}} \\ &= \frac{88.07 \text{ g}}{0.500 \text{ mol}} \\ &= 176.1 \text{ g/mol}\end{aligned}$$

$$\rightarrow 1 \text{ mol Vit. C} \times \frac{176.1 \text{ g Vit C}}{1 \text{ mol Vit C}} \times 0.409 = 72.04 \text{ g C}$$

← mass % C

$$72.04 \text{ g C} \times \frac{1 \text{ mol C}}{12.01 \text{ g C}} = 5.998 \text{ mol C} \approx 6 \text{ mol C} \quad \text{in 1 mol}$$

6 atoms C in 1 molecule

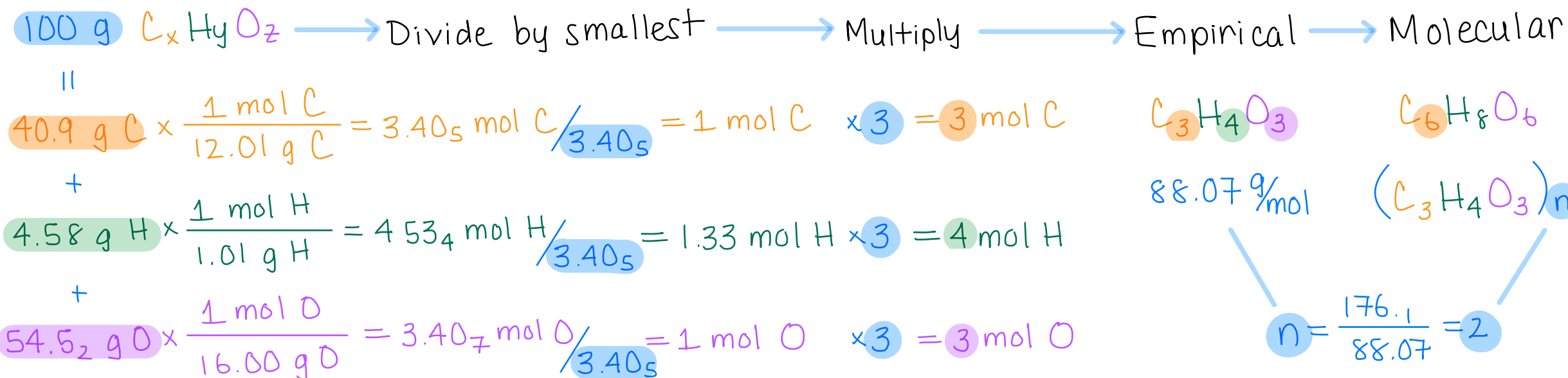
IN-CLASS QUESTION 4

multiple solutions!

Learning Objective(s): 3.5, 3.7

Vitamin C contains the elements C, H, and O. It is known to contain 40.9% C and 4.58% H by mass. The mass of 0.500 moles of vitamin C is 88.07 g. How many atoms of carbon are in a molecule of vitamin C?

Approach 2: Determine molecular formula, MM Vit C = 176.1 g/mol



IN-CLASS QUESTION 4

multiple solutions!

Learning Objective(s): 3.5, 3.7

Vitamin C contains the elements C, H, and O. It is known to contain 40.9% C and 4.58% H by mass. The mass of 0.500 moles of vitamin C is 88.07 g. How many atoms of carbon are in a molecule of vitamin C?

Approach 3: Compare mol C : mol Vit C assuming 100 g $C_xH_yO_z$, MM Vit C = 176.1 g/mol

$$\text{If } 100 \text{ g } C_xH_yO_z, \text{ then } 40.9 \text{ g C} \times \frac{1 \text{ mol C}}{12.01 \text{ g C}} = 3.405 \text{ mol C}$$

$$\text{and } 100 \text{ g } C_xH_yO_z \times \frac{1 \text{ mol } C_xH_yO_z}{176.1 \text{ g } C_xH_yO_z} = 0.5677 \text{ mol } C_xH_yO_z$$

$$\frac{3.405 \text{ mol C}}{0.5677} : \frac{0.5677 \text{ mol } C_xH_yO_z}{0.5677} = 6 \text{ mol C} : 1 \text{ mol } C_xH_yO_z$$